Historical Perspectives

At the Cutting Edge of the Impossible

A Tribute to Vladimir P. Demikhov

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Vladimir P. Demikhov (1916–1998) performed the world's first experimental intrathoracic transplantations and coronary artery bypass operation. His successes heralded the era of modern heart and lung transplantation and the surgical treatment of coronary artery disease. Even though he was one of the greatest experimental surgeons of the 20th century, his international isolation fueled speculation, suppositions, and myths. Ironically, his transplantation of a dog's head drew more publicity than did his pioneering thoracic surgical accomplishments, and he became an easy target for criticism. An account of Demikhov's life and work is presented herein. ***(Tex Heart Inst J 2009;36(5):453-8)***

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> *What I want and aim at is confoundedly difficult, and yet I do not think I aim too high. I want to do drawings which touch some people . . . I want to progress so far that people will say of my work, he feels deeply, he feels tenderly—notwithstanding my so-called roughness, perhaps even because of it. It seems pretentious to talk this way now, but this is the reason why I want to push on with all my strength. What am I in most people's eyes? A nonentity or an eccentric and disagreeable man—somebody who has no position in society and never will have, in short, the lowest of the low. Very well . . . then I should want my work to show what is in the heart of such an eccentric, of such a nobody. This is my ambition, which is, in spite of everything, founded less on anger than on love.*[1]
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> — Vincent van Gogh *(1853–1890)*

Vladimir P. Demikhov's success with experimental coronary artery surgery and intrathoracic organ transplantations heralded the modern era of heart and lung transplantation and the surgical treatment of coronary artery disease. However, Demikhov is known today mainly as a legendary surgeon who experimentally transplanted a second head onto a dog. His international isolation fueled speculation, suppositions, and myths. Although Demikhov's contributions to cardiothoracic surgery became internationally recognized during recent decades,[2-10] almost no details were known about him. Presented here is the biography of the man behind the legends—one of the greatest experimental surgeons of the 20th century.

Demikhov's Early Years

Vladimir P. Demikhov was born on 18 July 1916, into a family of peasants in the village of Yarizenskaia in Russia's Voronezh region. His father was killed during the Russian civil war. His mother was apparently a very determined woman: although she was herself minimally educated, she strove to provide a higher education for all 3 of her children.

In 1934, Demikhov left home in order to study biology at the University of Moscow. Upon arriving, he faced his first challenge: fulfilling the university's requirement that all newly admitted students present a photograph of themselves wearing a white shirt and a necktie. Demikhov had never had a white shirt or a tie, or the money to buy them. A kind photographer solved this problem for him by superimposing this apparel onto Vladimir's picture (Fig. 1).

Education, Enthusiasm, and Early Experiments

In 1937, Demikhov designed the first mechanical cardiac-assist device. Although it was too large to be installed inside the chest of a dog (the primary animal upon which Demikhov experimented), it could take over cardiac function for approximately

Fig. 1 Vladimir P. Demikhov (1916–1998), upon his admission to the University of Moscow in 1934 (photograph courtesy of Olga V. Demikhova).

5 hours. His experiments were the first ever in which circulation was maintained in an animal whose heart had been excised.[2]

Implantation of an artificial heart had appeared to be impossible. The pioneering nature and practical implications of this innovation were not appreciated, and yet Demikhov was not overly troubled. He was young, and full of energy, enthusiasm, and new ideas. Young men of that time dreamed about aviation, and Demikhov was no exception. He studied gliding and even took a flight, the recollection of which made him smile. A rumbling truck dragged his glider across a field as his friends ran behind. The glider took off, and it flew for a couple of minutes at about 2 meters above the field before it hit the ground and flipped over. When the disoriented Demikhov clambered out, his friends, all gasping for air, surrounded him. One of them finally managed to catch his breath and ask, "How was it in the sky?" Thus ended his short-term passion for airplanes. At that moment, he vowed to undertake activities that were more terrestrial.

Upon his graduation from the University of Moscow in 1940, Demikhov began working there as an assistant in the Department of Physiology. He transplanted a heart into the inguinal region of a dog. Not long thereafter, he realized that "because of its anatomical and physiological features, the heart can only function actively when it is transplanted into the thorax. If it is transplanted to the vessels of the neck or into the inguinal region, it cannot take an active part in the movement of the blood, and it is a neutral organ, living on the recipient's blood."[3,4]

World War II, and Demonstration of Character

His research was disrupted by World War II. After completing basic military training, Demikhov was accorded the rank of lieutenant. He served as a pathologist in a field evacuation hospital and saw, firsthand, all the horrors of war. His exemplary sense of honesty—a hallmark of his life—was challenged with potentially fatal consequences during this time. Years after the war, after having heard many tales about her father's honesty, his daughter asked, "Did you ever lie?" "Yes," he replied, "I lied a lot." Demikhov told her that the stress in the combat zone was overwhelming, and that many soldiers shot themselves in order to "escape" to the comparative refuge of the hospital. This was considered a war crime, the punishment for which was death. Demikhov was consulted as a forensic expert regarding such shootings. Although in most patients it was obvious to him that an injury was self-inflicted, he tried his best to attenuate the evidence, and thus he considered himself to have lied. His "lies" saved many lives. Demikhov knew all too well what fate would have befallen him had his falsehoods been revealed.

The Postwar Years

When Germany capitulated, Demikhov was in Berlin. The USSR declared war on Japan. Earlier in 1945, Demikhov and his unit had traveled by train halfway around the world, from Berlin to Harbin, China. At the year's end, Demikhov was able to return to Moscow. His siblings were working there: his brother as an accountant, and his sister as a biology teacher (Fig. 2).

In early 1946, experimenting on dogs, Demikhov performed intrathoracic transplantations of a heart, a lung, and the heart and lungs together—the first successful

Fig. 2 Vladimir Demikhov (right) with his brother Viacheslav (1913–1970) and sister Julia (1919–), 1946 (photograph courtesy of Olga V. Demikhova).

procedures of this kind that had been performed in any mammal.[7] Demikhov performed these transplantations without the use of cardiopulmonary bypass or hypothermia; instead, he relied heavily on speedy surgery and his self-designed technique of organ preservation during transfer. On 30 June 1946, a dog survived heterotopic heart–lung transplantation for 9.5 hours, marking Demikhov's first genuine success with this procedure. In 1969, Cooper[5] credited Demikhov with achieving transplantation of the heart and both lungs into an orthotopic locus, stating that Demikhov's "ingenious" technique "enabled the blood supply to the brain to be maintained continuously throughout the operation, with the exception of 2 to 3 minutes at the critical stage."

Demikhov began his 52-year marriage to his wife, Lia, in August 1946. Their only child, daughter Olga, was born on 16 July 1947.

Demikhov's experimental animals lived as long as 30 days after undergoing transplantation of both lungs, due in part to his preservation of the nerves of the diaphragm and normal function of abdominal organs after intrathoracic transplantation.

From 1947 into 1955, Demikhov conducted his experiments in Moscow at the Institute of Surgery (Fig. 3).[10] Alexander V. Vishnevsky directed the Institute. In the 1950s, a review committee of the Soviet Ministry of Health decided that Demikhov's work was unethical, and he was commanded to cease his research projects. However, Vishnevsky was surgeon-in-charge of the Soviet armed forces and was thus independent enough to be able to disobey the Ministry of Health and shelter Demikhov's research activities. Demikhov later worked at the Sechenov Medical Institute in Moscow (1955–1960) (Fig. 4) and with the Sklifosovsky Emergency Institute (1960–1986).

Fig. 3 *Vladimir Demikhov (standing, 4th from left), Alexander V. Vishnevsky (seated, 4th from left), and their colleagues at the Institute of Surgery in Moscow, 1953 (photograph courtesy of Olga V. Demikhova).*

In the early 1950s, Demikhov insisted that his mother come to live with his family. The 4 of them lived in 2 tiny rooms from 1954 until 1972, when his mother died at age 77. Besides his relatives, Demikhov often kept an experimental dog at home so that he could keep an eye on it all the time. Fortunately, he had a very supportive family (Fig. 5).

Fig. 4 *Vladimir Demikhov and one of his experimental dogs, in front of his laboratory at the Sechenov Medical Institute in Moscow (photograph courtesy of Olga V. Demikhova).*

Fig. 5 *Vladimir Demikhov with his wife Lia and daughter Olga, 1960s (photograph courtesy of Olga V. Demikhova).*

The Use of the Heart–Lung Preparation

Demikhov kept his donor heart–lung preparations viable during transfer by means of closed-circuit circulation. Blood from the left ventricle was pumped into the aorta; then, through the coronary vessels that supplied the myocardium, it passed into the right atrium, the right ventricle, and the lungs, where the blood was reoxygenated and returned to the left atrium.[3,4,11] Demikhov based his design of this heart–lung preparation on the original work of Ivan P. Pavlov. In 1886, Pavlov and his associate N.J. Chistovich had designed the first preparation in which the heart and the lungs were maintained alive as they functioned on their own power. This preparation was subsequently useful in research into the pharmacologic action of various drugs.[12] In 1912, Knowlton and Starling described a modified heart–lung preparation (Starling's preparation).[13] Demikhov simplified this preparation and used it in the early 1950s, saying that "in the future, when the transplantation of the human heart and lungs is a practical possibility, the transfer of the organ in a functioning state will be facilitated by the use of this preparation."[4] Subsequently, Robicsek and colleagues[14] reported their modification of the preparation, which would enable clinically applicable preservation of the heart and lungs. In 1987, Hardesty and Griffith[15] reported their successful use of modified, autoperfused heart–lung preparation in clinical transplantation. In the 1950s, however, it had seemed impossible that heart–lung preparation would ever achieve any clinical use.

The First Coronary Bypass Operation

Experimenting on a dog, Demikhov performed the first successful coronary bypass operation, on 29 July 1953. To achieve the coronary anastomosis, Demikhov used his personal adaptation of Payr's technique, which had originally been described in 1900. Four dogs survived longer than 2 years, and anastomotic patency was proved in all.[3,4] Demikhov explored possible clinical application by experimenting on cadavers and baboons.

Because the widespread clinical application of coronary bypass surgery seemed impossible in the early 1960s, Demikhov's experimental work of the previous decade was initially regarded by many as impractical and quite eccentric.[16] Undaunted by such criticism, V.I. Kolesov in Leningrad undertook further experiments. From 25 February 1964 through 9 May 1967, the department of surgery that Kolesov directed was the only place on earth where coronary bypasses were performed. Kolesov acknowledged Demikhov's pioneering contributions in his first publication[17] and in many thereafter.

Head Transplantation and the Firestorm of Controversy

In 1954, Demikhov performed canine head transplantation (Fig. 6). The maximal survival of any animal was 29 days.[3,4,8] Ironically, the news of this pioneering surgery spread around the world far more rapidly than had any reports of Demikhov's earlier experiments. The operation raised many eyebrows and even more ethical questions: newspapers everywhere were imbued with discussions of various perspectives, ethics arguments, and controversies. In 1997, one author[18] recalled that in 1962 he had been "sitting in the surgeon's changing room when a report of Demikhov's head transplant appeared in the Cape Argos newspaper. The news was conveyed to Christiaan Barnard, who was clearly put out. He stormed out with the retort that 'anything those Russians can do, we can do, too.' The same afternoon, Barnard transplanted the head of a dog onto a recipient dog, which survived for several days. Animal-rights protectors were incensed, and the medical students built a papier-mâché 2-headed dog for their RAG [raising and giving] parade. Barnard had already walked the tightrope between genius and vulgarity."

Fig. 6 *Experimental dog with two heads (photograph courtesy of Olga V. Demikhova).*

The head transplantation by Demikhov was arguably the most controversial experimental operation of the 20th century. It fomented waves of indignation in medical circles, and Demikhov—whose experiments were always an easy target for criticism—was accused of being a charlatan. Why did he perform head transplantation? There will never be a clinical application of this procedure in any modification. This is simply impossible. This is the only operation of Demikhov's that will never find its clinical application. Or . . . or will it?

Belated Recognition

Due to the Cold War, Demikhov rarely appeared outside the Soviet Union. In 1958, he gained peer recognition after he demonstrated experimental transplantations in Leipzig, Germany (Fig. 7). On 16 September 1960, Demikhov was given membership in the Royal Scientific Society of Uppsala, Sweden. That same year, his monograph[4] was published in Moscow, and it was translated and further published (in New York, 1962[3];

in Berlin, 1963; and in Madrid, 1967). This was the world's first book that discussed intrathoracic transplantation. A year after the first translation, James Hardy performed the first transplantation of a human lung. Other surgeons became eager to visit Demikhov's research facilities, but only a few succeeded. One of those few was Christiaan Barnard, who visited Moscow as a tourist in 1962 and was able to divert himself from the planned itinerary. In 1997, Barnard wrote to me that Demikhov "was certainly a remarkable man, having done all the research before extracorporeal circulation. I have always maintained that if there is a father of heart and lung transplantation, then Demikhov certainly deserves this title."

In April 1989, the International Society for Heart and Lung Transplantation was "privileged to present the first Pioneer Award to Professor Demikhov of the Soviet Union for his leadership role in the development of intrathoracic transplantation and the use of artificial hearts."[9] At that year's meeting of the Society, in Munich, Germany, Demikhov personally received the award (Fig. 8) in front of an appreciative audience that included his daughter Olga. This tribute to his achievements, belated though it was, partially made amends for his decades of labor in comparative obscurity.

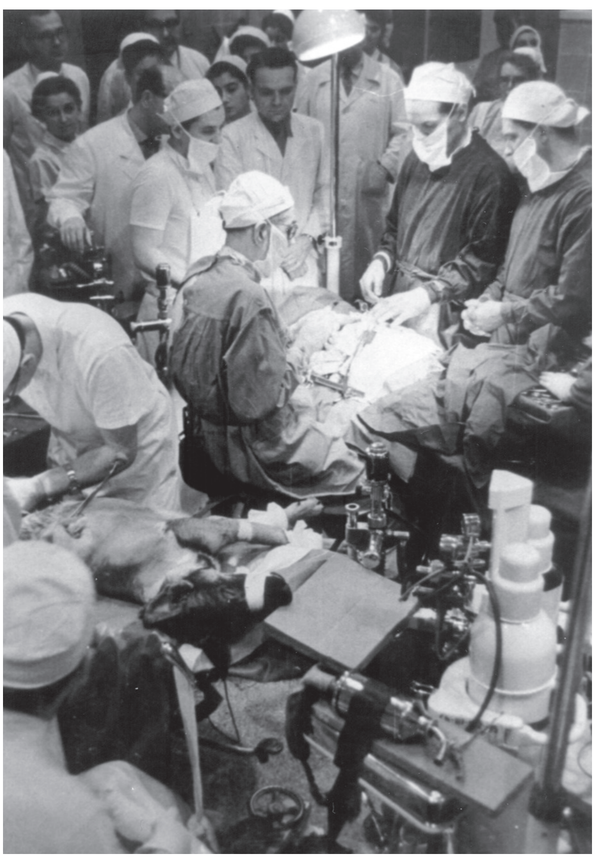

Fig. 7 *Vladimir Demikhov performs heart–lung transplantation on 16 December 1958 in Leipzig, Germany (photograph courtesy of Olga V. Demikhova).*

Final Years, and Death

Stroke with memory loss made it impossible for Demikhov to update his 1960 monograph, as he had desired (Fig. 9). In April 1998, Demikhov was hospitalized

Fig. 8 *Christian Cabrol (left) presents the Pioneer Award of the International Society for Heart and Lung Transplantation to Vladimir Demikhov on 25 April 1989 in Munich, Germany (photograph courtesy of Olga V. Demikhova).*

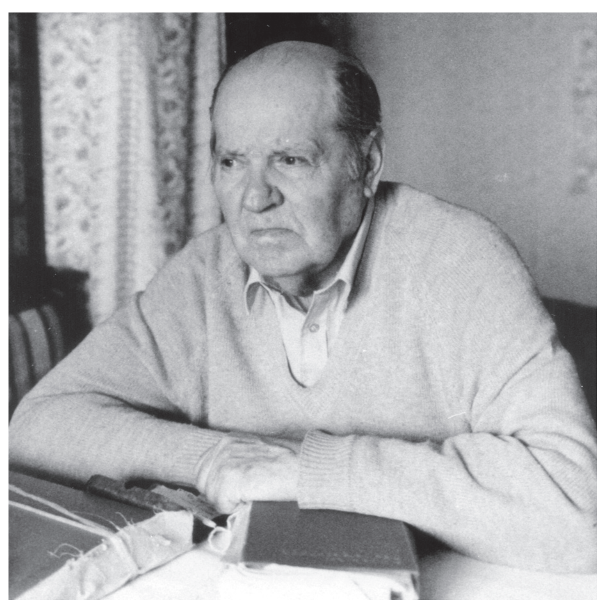

Fig. 9 *Vladimir Demikhov with his unpublished monograph revision, January 1998 (photograph courtesy of Olga V. Demikhova).*

with recurrent stroke. He was paralyzed and developed pneumonia. His wife Lia died on 11 July, and he never recovered from this loss. Although Demikhov was discharged after some months in the hospital, he remained bedridden. He died in his small apartment outside Moscow on 22 November 1998. The world of medicine as Demikhov left it was almost unrecognizable from the world that he had entered—most of his ideas were now a routine part of clinical practice.

Demikhov's Legacy

The paradox surrounding Vladimir Demikhov was that he was always exploring the unknown, far ahead of the times. His work produced much anxiety in lay and medical circles. Public and scientific attitudes toward his work changed many times. Immediate judgment was difficult—too new and unconventional were his experiments. Some of his contemporaries considered his work to be on the cutting edge of surgical research, and they were eager to meet him. Others deemed his work fanciful or vulgar and were reluctant to admit any association with him. Regardless of the attitudes toward Demikhov's accomplishments, his work touched some people and made them believe in what seemed impossible. Only upon the passage of time can one appreciate the true impact of his innovations.

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